

EXP-011 — Controlled Query-Side Retrieval

Production RAG v1 · evaluation-first · n = 20 questions / 22 evidence spans · document side frozen

STATUS — HYPOTHESIS FALSIFIED, WITH AN INSTRUCTIVE TWIST

Every query transformation made the *system* worse, and **not one case was rescued in any cell**. But the same transformations made each retriever **better in isolation**. The hybrid's strength was never query quality — it was the two retrievers failing on different questions, and "improving" the query destroyed that.

0

cases rescued
across all four cells

+0.100

BM25 gain from normalization
on its own

+4 → +1

fusion bonus collapse
raw → normalized

3×

retrieval work for cell E
to lose two cases

1. Why EXP-011 exists

EXP-010 closed the document side: truncation eliminated completely (23.22% of chunks → 0%, token coverage 0.7610 → 1.0000) and retrieval moved $\Delta 0.000$, with **21 of 22 answers already visible** to the encoder. What remains is a ranking problem. The query had been a raw user-question string since EXP-000 — the one major variable never tested.

2. Frozen system and leakage controls

Corpus, chunks, **stored document embeddings** (fingerprint `bd95feaeacf98559`), transformer, tokenizer, 512-token window, BM25 parameters, cosine, exact search, RRF pool 50 / `rrf_k` 60 / `top_k` 10, evidence anchors and questions — all unchanged. **Only the query text differed.**

A transform receives only the raw question string. Its module imports *only* `re` and `dataclasses` — no project import at all — and tests assert that its executable code names no evaluation symbol and hardcodes no golden question or evidence path.

3. The transformations

```
raw      : What is the default value of max_tokens in the Anthropic Messages API?
normalized : default value max_tokens Anthropic Messages API
structured : max_tokens Anthropic Messages API default value
```

A held-out probe, not a golden question. Identifiers, numbers, status codes and product names the user wrote are protected; provider names are never invented.

4. Reproduction gate

Cell A reproduced the frozen hybrid exactly — **0.775 / 15-of-20 / 17-of-22: PASS**.

5. Results

cell	views	macro R	full	spans@10	doc R	MRR	a@10	a@30	a@300	calls
A — raw query (control)	1	0.775	15/20	17/22	0.925	0.449	5	4	2	2
B — normalized	1	0.700	13/20	15/22	0.900	0.371	7	4	2	2
C — raw + normalized	2	0.750	14/20	16/22	0.950	0.405	6	4	2	4
D — structured	1	0.300	5/20	7/22	0.725	0.092	15	11	8	2
E — raw + normalized + structured	3	0.700	13/20	15/22	0.950	0.319	7	4	1	6

6. Paired movement — no cell rescued anything

comparison	Δ	rescued	regressed	net	cases lost
A->B	-0.075	0	2	-2	AN-002, OA-006
A->C	-0.025	0	1	-1	OA-006
A->D	-0.475	0	10	-10	AN-002, AN-004, AN-005, OA-001, OA-002, OA-003, OA-004, OA-005, OA-006, OA-008
A->E	-0.075	0	2	-2	OA-002, OA-006

Zero rescues across every cell and every case — a stronger result than the aggregate deltas. There is no subset of questions for which query rewriting helped.

7. The twist: the retrievers individually got better

retriever	raw	normalized	Δ	structured	Δ
bm25	0.475 (9/20)	0.575 (11/20)	+0.100	0.125 (2/20)	-0.350
transformer	0.575 (11/20)	0.625 (12/20)	+0.050	0.300 (5/20)	-0.275

Normalization improved **both** retrievers alone — BM25 by two cases, the transformer by one. Yet fusing the two improved retrievers produced a worse system than fusing the two unimproved ones.

8. Mechanism: the fusion bonus collapsed

query view	BM25 alone	transformer alone	best component	fused	fusion bonus
raw	9/20	11/20	11	15	+4 cases
normalized	11/20	12/20	12	13	+1 cases
structured	2/20	5/20	5	5	+0 cases

WHAT RRF WAS ACTUALLY BUYING

Fusion adds value in proportion to how much the retrievers still *disagree usefully*. On the raw query it is worth **four cases** beyond its best component. Normalizing pushed both retrievers toward the same evidence and the bonus fell to **one** — more than cancelling the individual gains.

A supporting proxy agrees but only weakly, and is reported as such: mean Jaccard overlap between the retrievers' top-50 rose 0.134 → 0.154 → 0.187, while shared top-10 chunks stayed flat (2.5 → 2.4). The fusion-bonus table is the solid evidence.

9. The design rule earned its place

The preregistration forbade ever replacing the user's query, and that rule bounded the damage: B (normalized alone) loses two cases, C (raw + normalized) loses one. Keeping the original halved the loss. E, which adds a third view, is worse than C — a further view is not free even when the original is retained.

10. AN-003

cell	evidence rank	document rank
A — raw query (control)	absent@300	3
B — normalized	absent@300	6
C — raw + normalized	absent@300	4
D — structured	absent@300	3
E — raw + normalized + structured	absent@300	2

No query representation moved it. The pattern is now unmistakable: the **right document ranks 2nd–6th in every configuration** while the chunk carrying the answer never enters the top 300. Not a query problem, not a visibility problem — a **within-document chunk-ranking** failure, surviving its ninth experiment.

11. Candidate depth and the reranker gate

cell	1–10	11–30	31–50	51–100	101–300	absent	ceil@30	ceil@50	ceil@100
A — raw query (control)	17	1	1	1	0	2	0.818	0.864	0.909
C — raw + normalized	16	2	2	0	0	2	0.818	0.909	0.909
E — raw + normalized + structured	15	3	1	1	1	1	0.818	0.864	0.909

Query expansion did **not** raise the perfect-reranker ceiling — 0.909 at pool 100 for A, C and E alike. Cell E pulls one span out of "absent" into the 101–300 band, which is real but not ceiling-changing. A reranker would still chase 0.909 against 0.775 already delivered, and would have to be perfect to collect it.

12. Cost

Cell E costs **three times** the retrieval work (6 calls per question vs 2) to lose two cases. Wall-clock latency is *not* comparable across cells here — A ran first and absorbed cold-cache and warm-up costs, which is why it shows the highest total despite the fewest calls. The call count is the honest measure. The query-embedding cache served 63.1% of requests.

13. Limitations

- $n = 20 / 22$ spans; one case is 5 percentage points. No significance claims.
- Two deterministic transforms were tested, not the space of query rewriting. An LLM rewriter might behave differently — though the mechanism found here predicts it would also erode complementarity if it made both retrievers agree.
- The fusion bonus is a case count at $n = 20$; ± 1 case is noise, but the $+4 \rightarrow +1 \rightarrow +0$ progression is monotone across three independent views.
- **EXP-NULL remains BLOCKED** — no project generation credential, so there is still no measured no-retrieval floor beneath any of these numbers. EXP-011F (LLM rewriting) was not run for the same reason.

14. Was query formulation a real bottleneck?

No — not in the way the hypothesis proposed. Rewriting cannot rescue a single case, and the raw question is already the best input to the hybrid.

But the finding is more specific than "queries are fine". Query form **does** change individual retriever quality materially — BM25 gained two cases from filler removal alone. What it cannot do is improve the *system*, because this system's performance comes from retriever disagreement rather than either retriever's absolute quality. **Optimising the components made the ensemble worse.** That is a result about ensembles, not about queries, and it was only visible because the components were measured separately from the fusion.

15. What the evidence justifies next

1. **Stop rewriting queries.** Preregistered Outcome D. Zero rescues in four configurations is not a near miss.
2. **Treat complementarity as the quantity to protect.** Future changes should report the fusion bonus, not just aggregate recall — a change that improves both retrievers can still degrade the system.
3. **The remaining failure is within-document chunk ranking.** AN-003 is the clean case: right document at rank 2, answer chunk never in the top 300.
4. **A reranker is still not justified** on ceiling grounds, and EXP-011 did not move that ceiling.
5. **EXP-NULL remains the most valuable unblocked experiment.**

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